

Logic Design and Switching Theory

Lecture – 11

Subtitle: Counters

by
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Previous

Latches

Flipflops

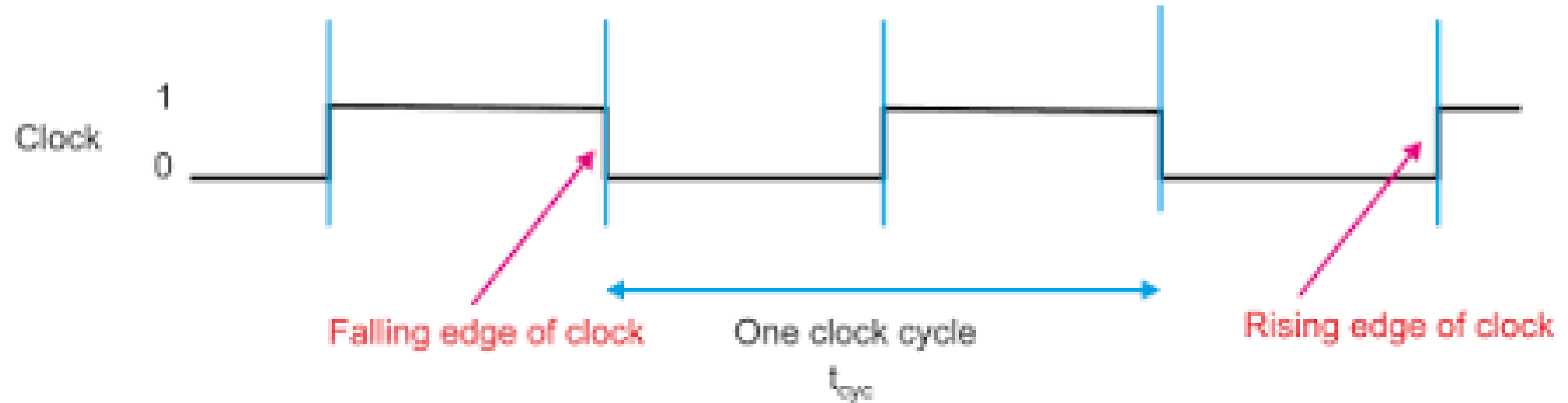
Today's Topic

Counters

- Asynchronous
- synchronous

Revision of Latches and Flipflops

Clock signal



Conversions of Flip flops

2. T flipflop to JK flipflop

Excitation table of T flipflop

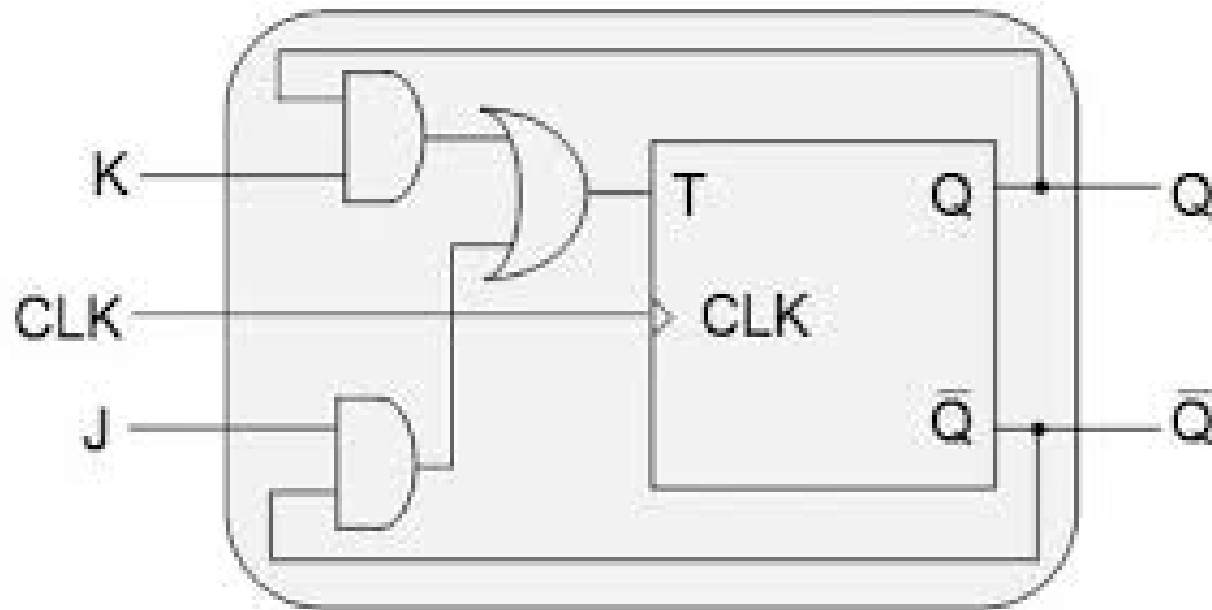
Q(current)	Q(next)	T
0	0	0
0	1	1
1	0	1
1	1	0

J	K	Q_N	Q_{N+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

	00	01	11	10
0				
1				

Conversions of Flip flops

2. T flipflop to JK flipflop



Conversions of Flip flops

3. JK flipflop to SR flipflop

Excitation table of JK flipflop

Q(current)	Q(next)	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

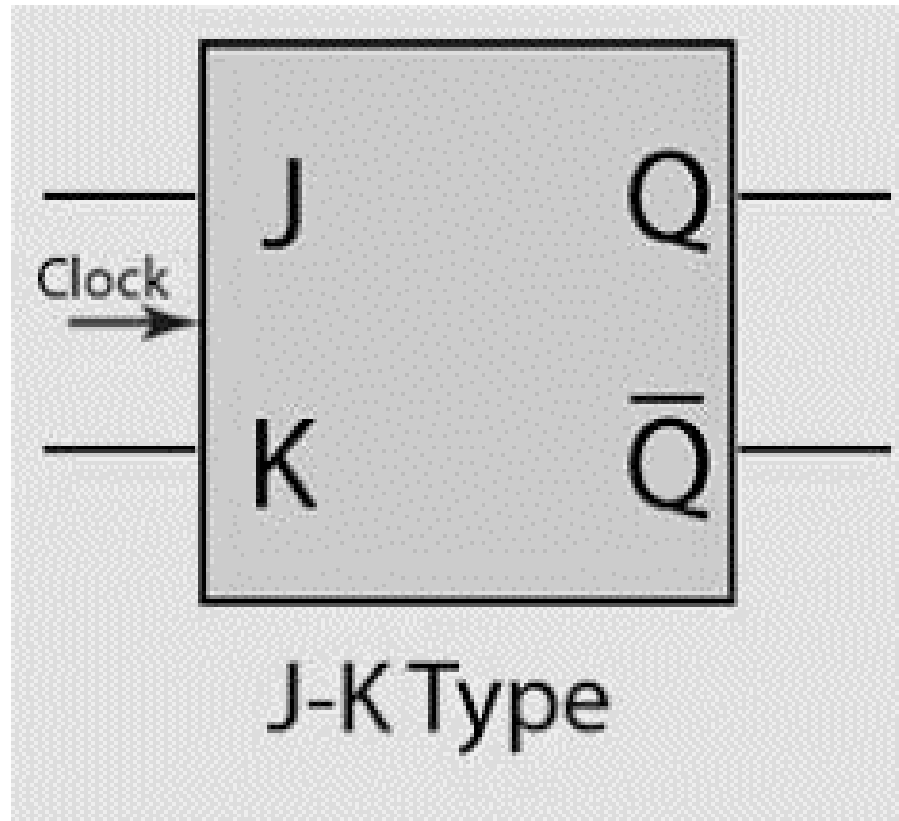
Input s		Present State	Next State	Flip Flop Input	
S	R	Q _n	Q _{n+1}	J	K
0	0	0	0		
0	0	1	1		
0	1	0	0		
0	1	1	0		
1	0	0	1		
1	0	1	1		
1	1	0	X		
1	1	1	X		

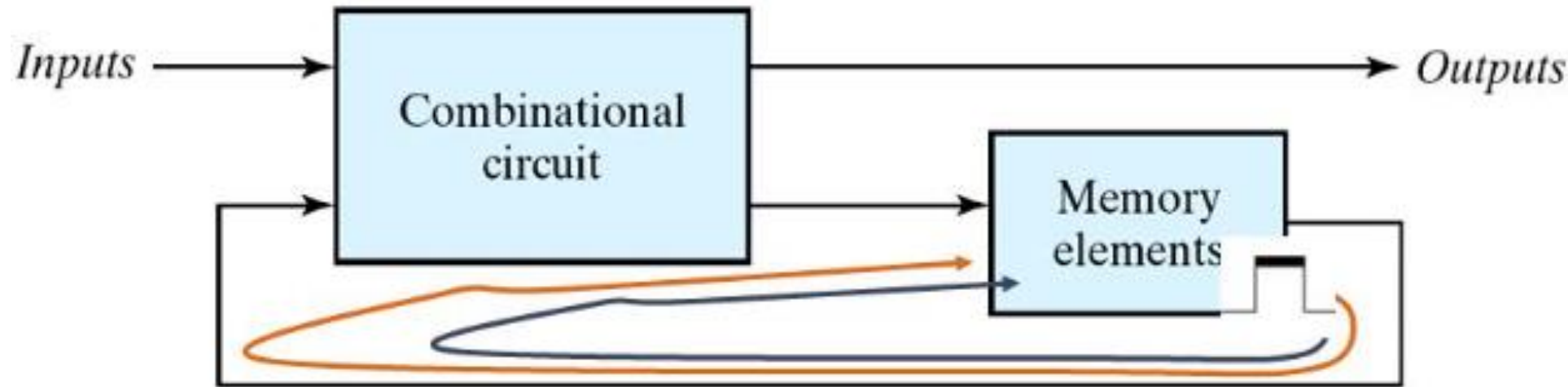
	00	01	11	10
0				
1				

	00	01	11	10
0				
1				

Conversions of Flip flops

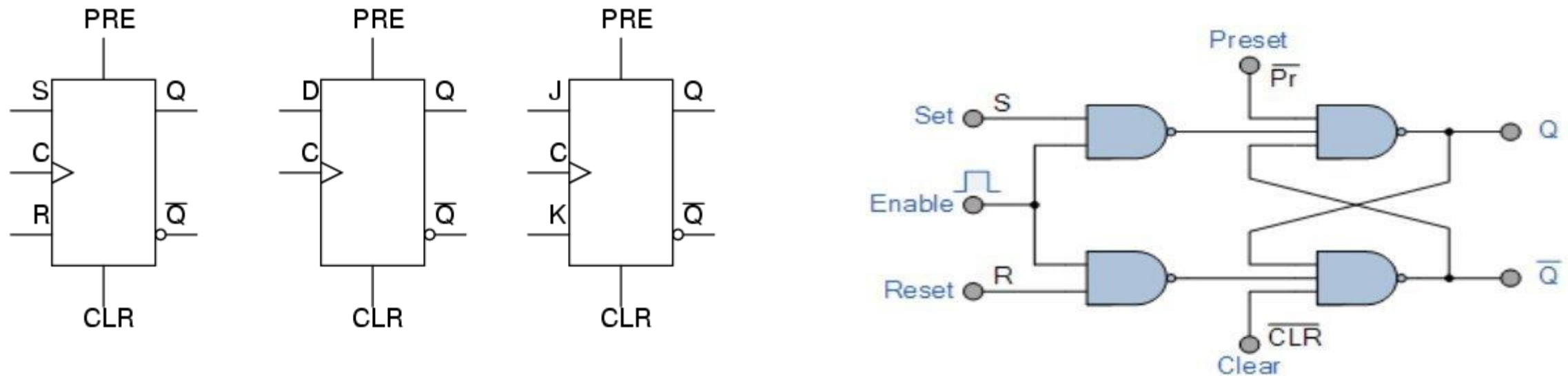
3. JK flipflop to SR flipflop





- If level-triggered flip-flops are used
 - the feedback path may cause instability problem (since the time interval of logic-1 is too long)
 - multiple transitions might happen during logic-1 level
- Edge-triggered flip-flops
 - the state transition happens only at the edge
 - eliminate the multiple-transition problem

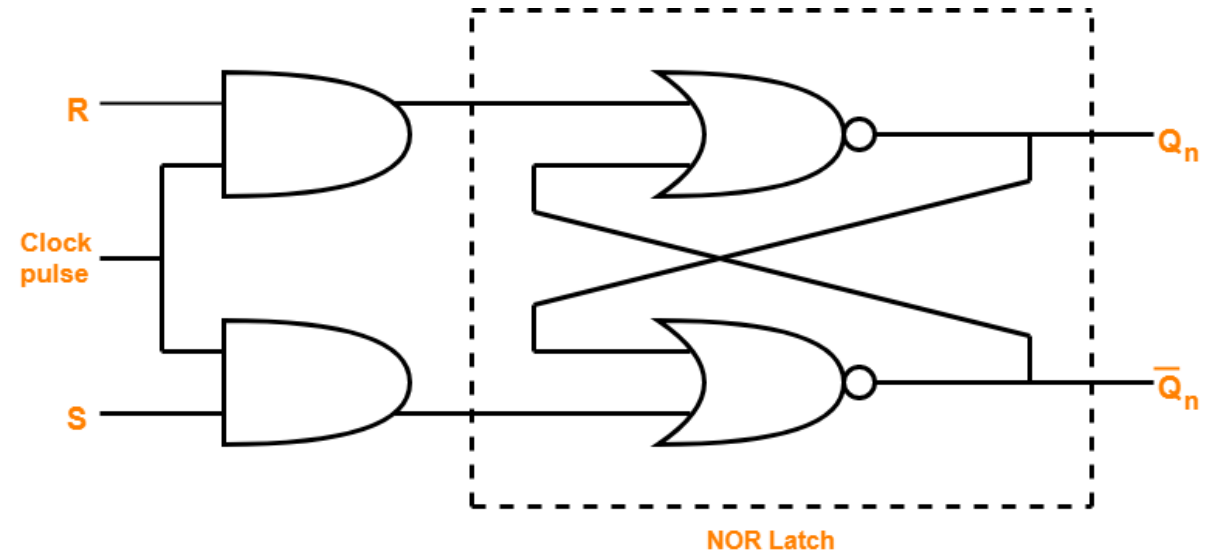
Asynchronous Pins (Preset & Clear)



- *Asynchronous* inputs on a flip-flop have control over the outputs (Q and not-Q) regardless of clock input status.
- These inputs are called the *preset* (PRE) and *clear* (CLR). **The preset input drives the flip-flop to a set state while the clear input drives it to a reset state.**
- It is possible to drive the outputs of a J-K flip-flop to an invalid condition using the asynchronous inputs, because all feedback within the multivibrator circuit is overridden.

SR Flipflop

$$Q_{n+1} = S + Q_n R'$$



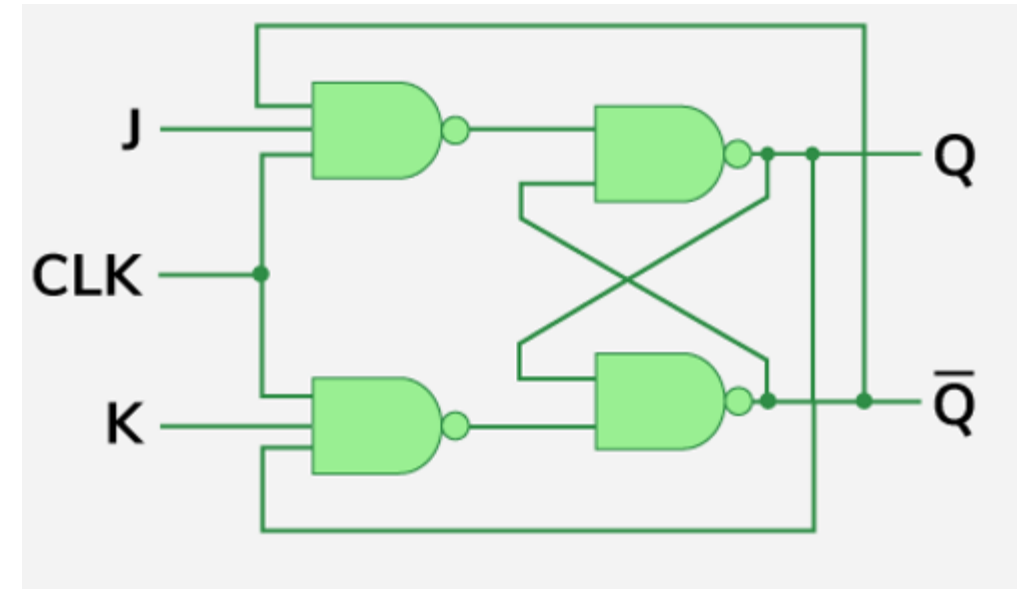
SR Flip Flop Using NOR Latch

S	R	Q _{n+1}
0	0	Q _n
0	1	0
1	0	1
1	1	Indeterminate

Q _n	Q _{n+1}	S	R
0	0	0	0
0	1	1	0
1	0	0	1
1	1	0	0

JK Flipflop

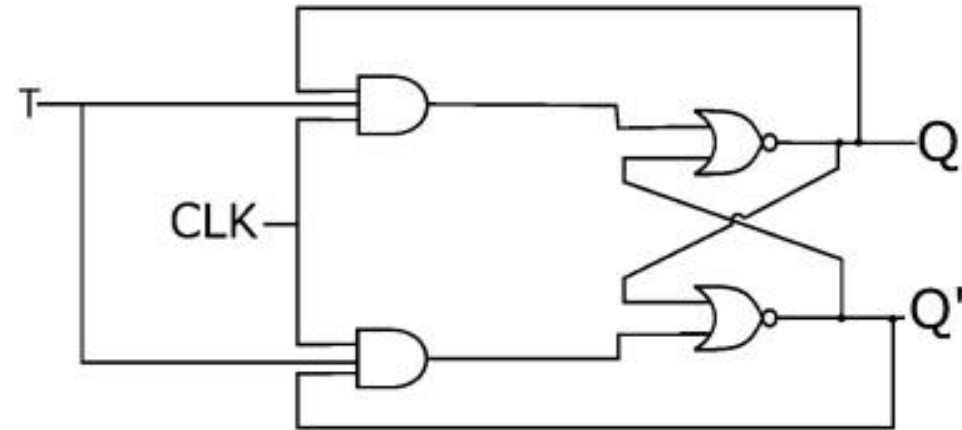
$$Q_{n+1} = \bar{K}Q_n + J\bar{Q}_n$$



J	K	Q(next)
0	0	Q (NC)
0	1	0 (Reset)
1	0	1 (set)
1	1	Q' (Toggle)

Q _n	Q _{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

T Flipflop

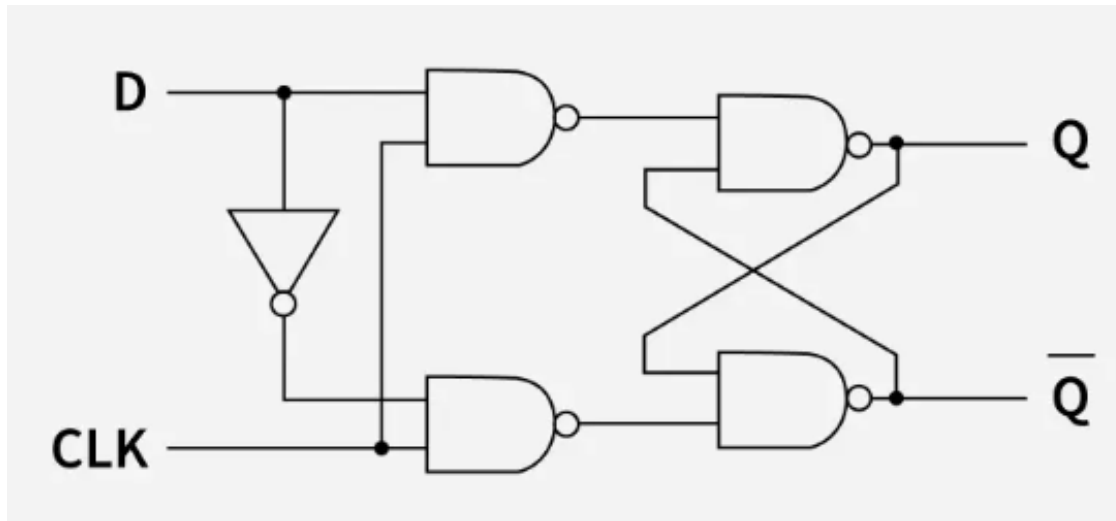


$$Q(n+1) = TQn' + T'Qn = T \oplus Qn$$

T	Q(next)
0	Q
1	Q'

Q(current)	Q(next)	T
0	0	0
0	1	1
1	0	1
1	1	0

D Flipflop

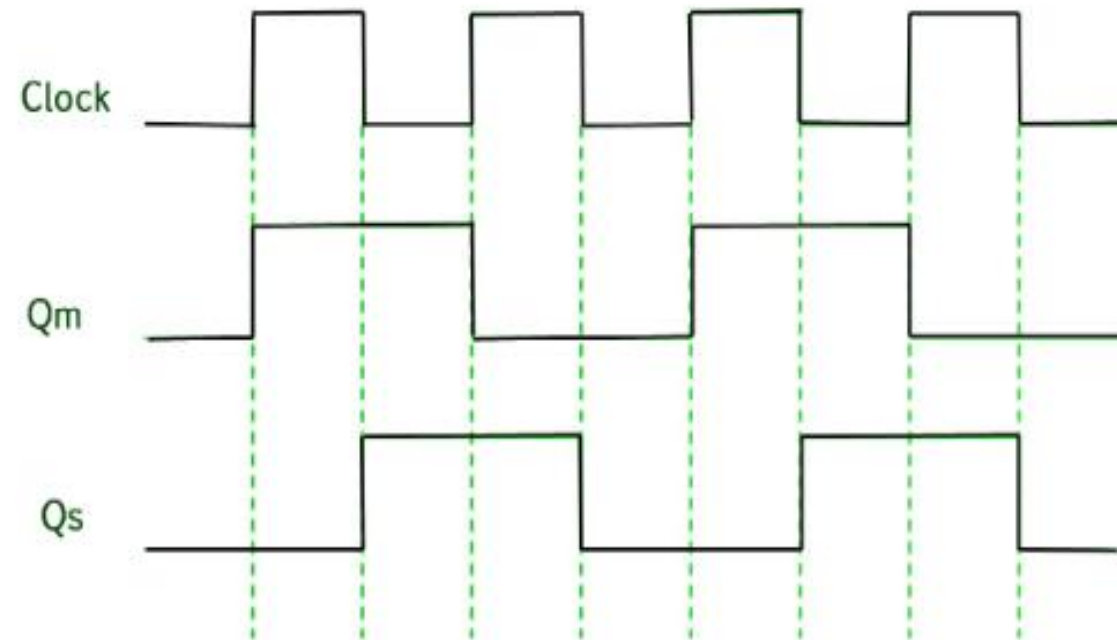
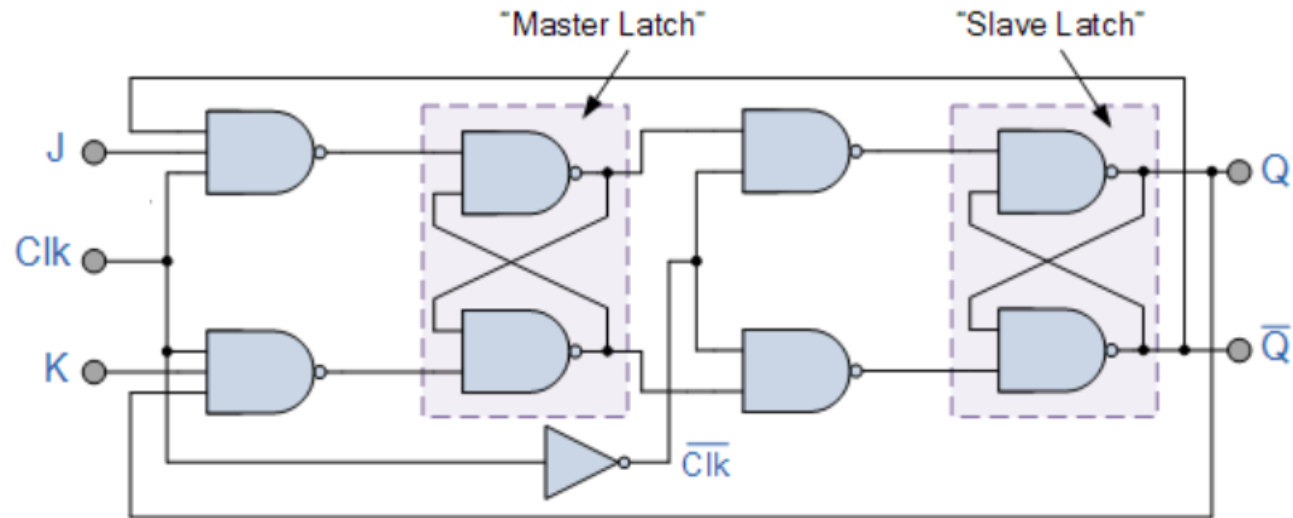


$$Q_{(n+1)} = D$$

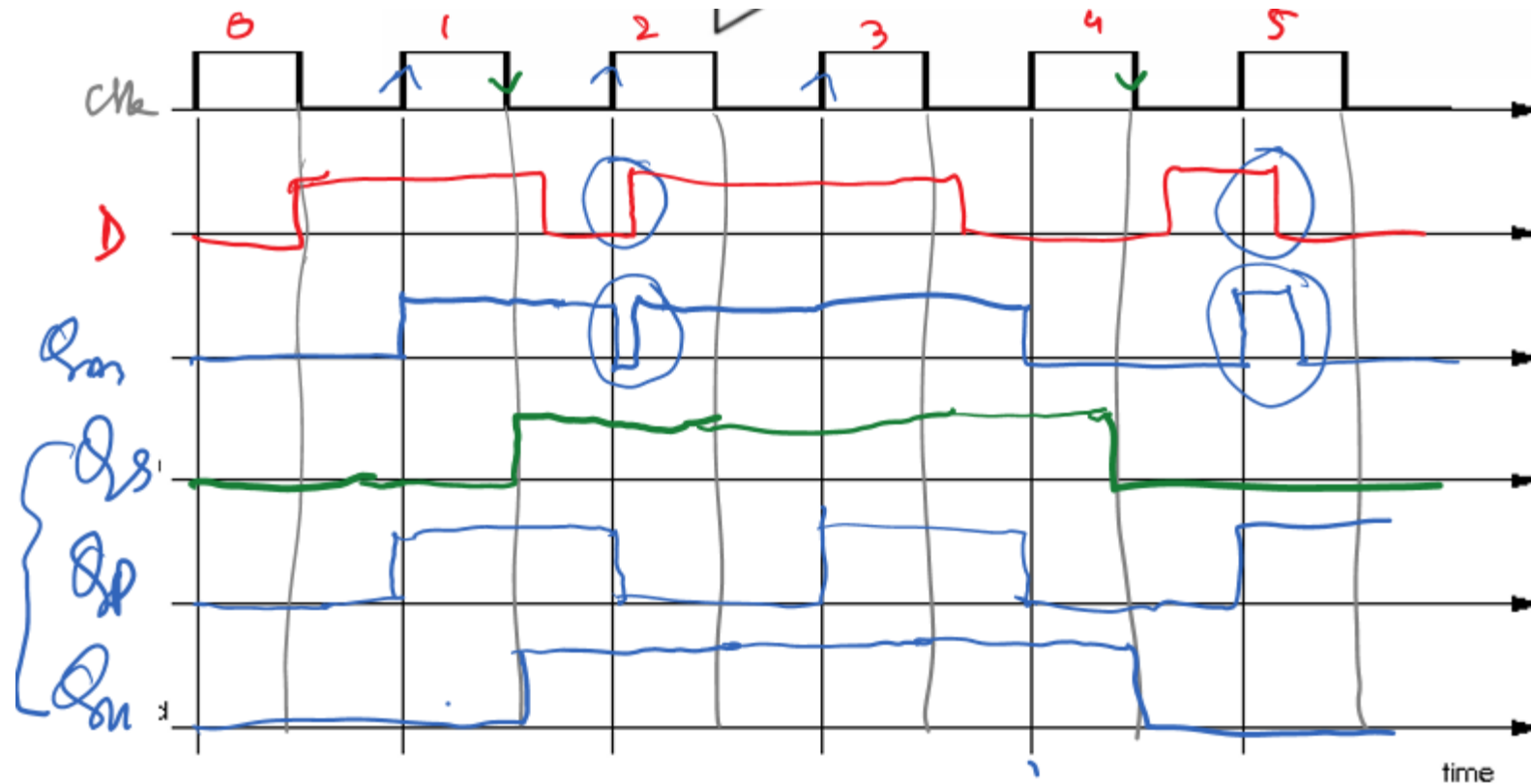
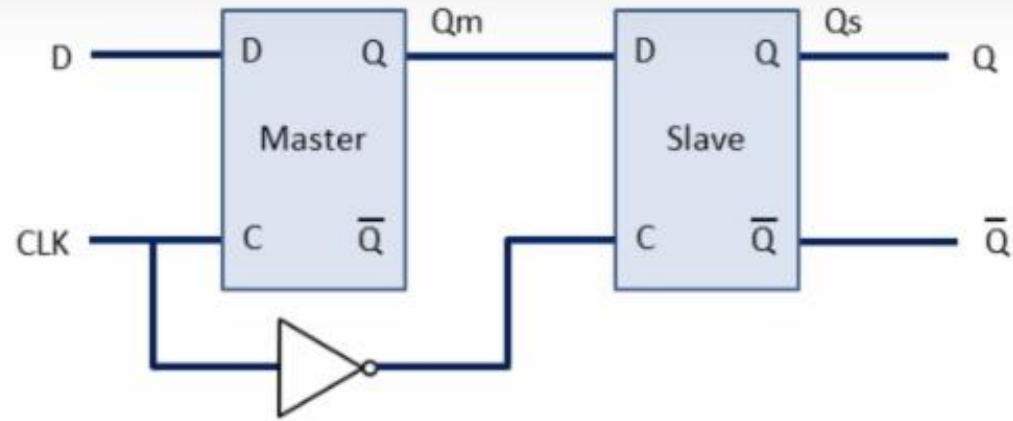
D	Q(next)
0	0
1	1

Q(n)	D(n)	Q(n+1) (Next)
0	0	0
0	1	1
1	0	0
1	1	1

Master-slave JK Flipflop



Master-slave D Flipflop



Counters

Counters

The combination of flipflops that perform counting operation is known as counters.

A Counter is a device which stores (and sometimes displays) the number of times a particular event or process has occurred, often in relationship to a clock signal. Counters are used in digital electronics for counting purpose, they can count specific event happening in the circuit.

Counters are sequential circuit that count the number of pulses can be either in binary code or BCD form. The main properties of a counter are timing , sequencing , and counting.

Counters

Asynchronous - Each flip-flop is triggered by the output of the previous one. Simple but slower because the signal "ripples" through each stage.

Synchronous - All flip-flops are triggered at the same time by the clock signal. Faster and more reliable at higher frequencies.

Up Counter - Counts up (0, 1, 2, 3...).

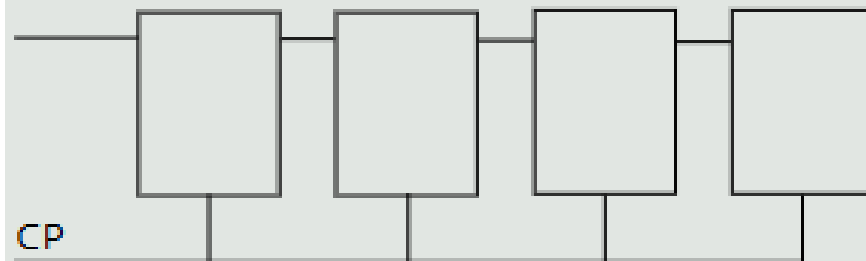
Down Counter - Counts down (9, 8, 7...).

Mod-N counter - Counts up to N-1 and then resets to zero. Example: A mod-10 counter counts 0–9 and then resets to 0 (used in digital clocks)

Ring counter - A circular shift register with only one '1' circulating among zeros.

Johnson Ring counter - A modified ring counter where the output of the last flip-flop is inverted and fed back to the input of the first.

Synchronous counter



Basic diagram

It have a non-sequential mode of counting (different state of counting 1 - 4 - 5- 7)

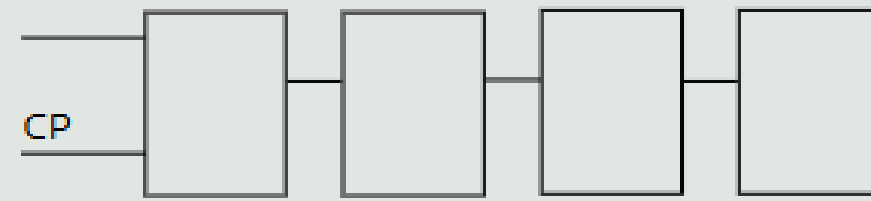
Ripple and Johnson counter

Faster

Complex circuit

Cost is high

Asynchronous counter



Basic diagram

It have a sequential mode of counting (different state of counting 1 - 2 - 3 - 4 OR 4 - 3 - 2 - 1)

Ripple counter

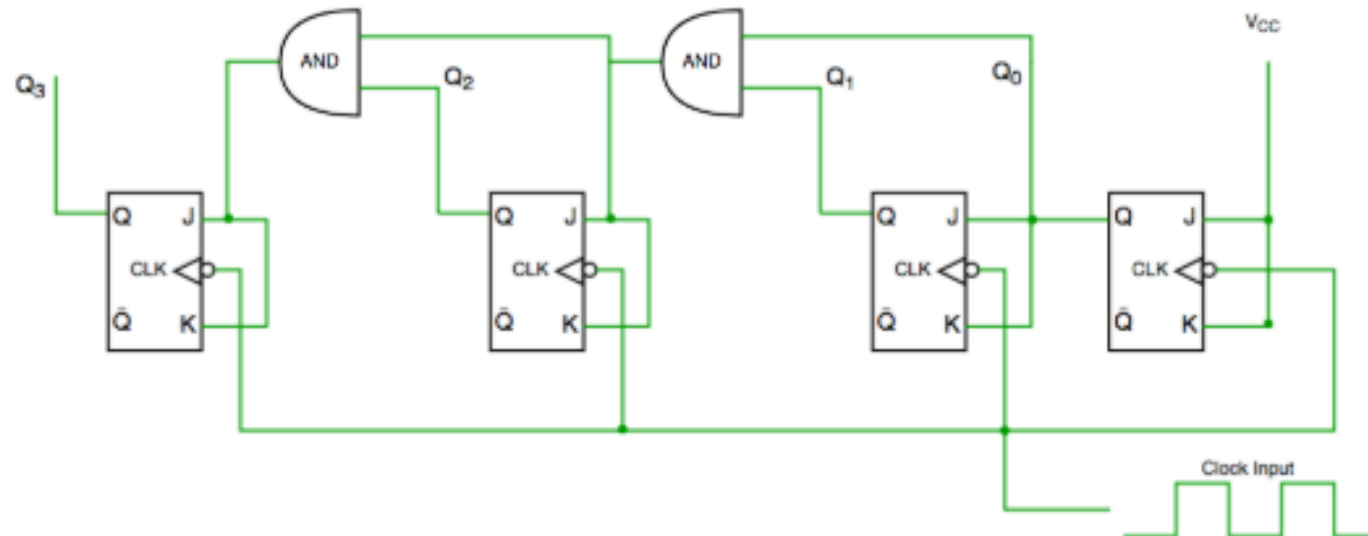
Slower

Simple circuit

Cost is low

Synchronous Counters

Unlike the asynchronous counter, synchronous counter has one global clock which drives each flip flop so output changes in parallel. The one advantage of synchronous counter over asynchronous counter is, it can operate on higher frequency than asynchronous counter as it does not have cumulative delay because of same clock is given to each flip flop. It is also called as parallel counter.



Synchronous Counters

2-Bit Synchronous Binary Counter

Working

- The operation of a J-K flip-flop synchronous counter is as follows:
First, assume that the counter is initially in the binary 0 state; that is, both flip-flops are RESET. When the positive edge of the first clock pulse is applied, FF0 will toggle and Q0 will therefore go HIGH. What happens to FF1 at the positive-going edge of CLK1? To find out, let's look at the input conditions of FF1. Inputs J1 and K1 are both LOW because Q0, to which they are connected, has not yet gone HIGH. Remember, there is a propagation delay from the triggering edge of the clock pulse until the Q output actually makes a transition. So, $J = 0$ and $K = 0$ when the leading edge of the first clock pulse is applied. This is a no-change condition, and therefore FF1 does not change state

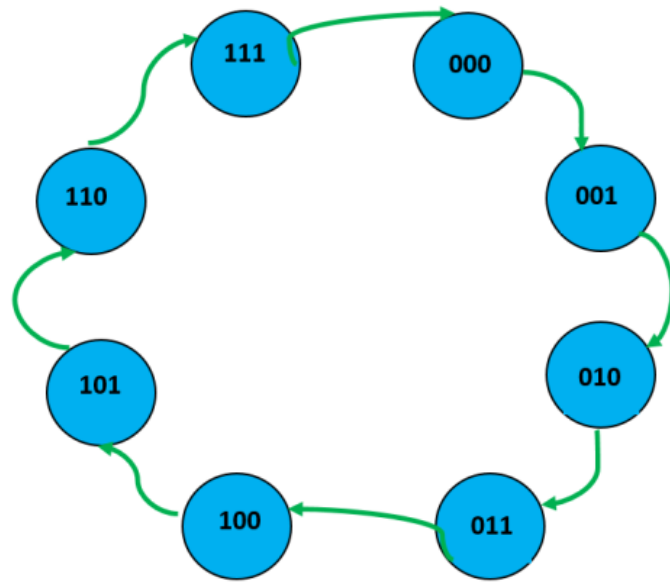
2-Bit Synchronous Binary Counter

Working

After CLK1, $Q_0 = 1$ and $Q_1 = 0$ (which is the binary 1 state). When the leading edge of CLK2 occurs, FF0 will toggle and Q_0 will go LOW. Since FF1 has a HIGH ($Q_0 = 1$) on its J_1 and K_1 inputs at the triggering edge of this clock pulse, the flip-flop toggles and Q_1 goes HIGH. Thus, after CLK2, $Q_0 = 0$ and $Q_1 = 1$ (which is a binary 2 state). The timing detail for this condition is shown in Figure 9-13(b).

When the leading edge of CLK3 occurs, FF0 again toggles to the SET state ($Q_0 = 1$), and FF1 remains SET ($Q_1 = 1$) because its J_1 and K_1 inputs are both LOW ($Q_0 = 0$). After this triggering edge, $Q_0 = 1$ and $Q_1 = 1$ (which is a binary 3 state). The timing detail is shown in Figure 9-13(c).

3-bit Synchronous Counter using JK FF



3-bit Synchronous Counter using JK FF

Counter State	Q_2	Q_1	Q_0
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

3-bit Synchronous Counter using JK FF

	00	01	11	10
0				
1				

	00	01	11	10
0				
1				

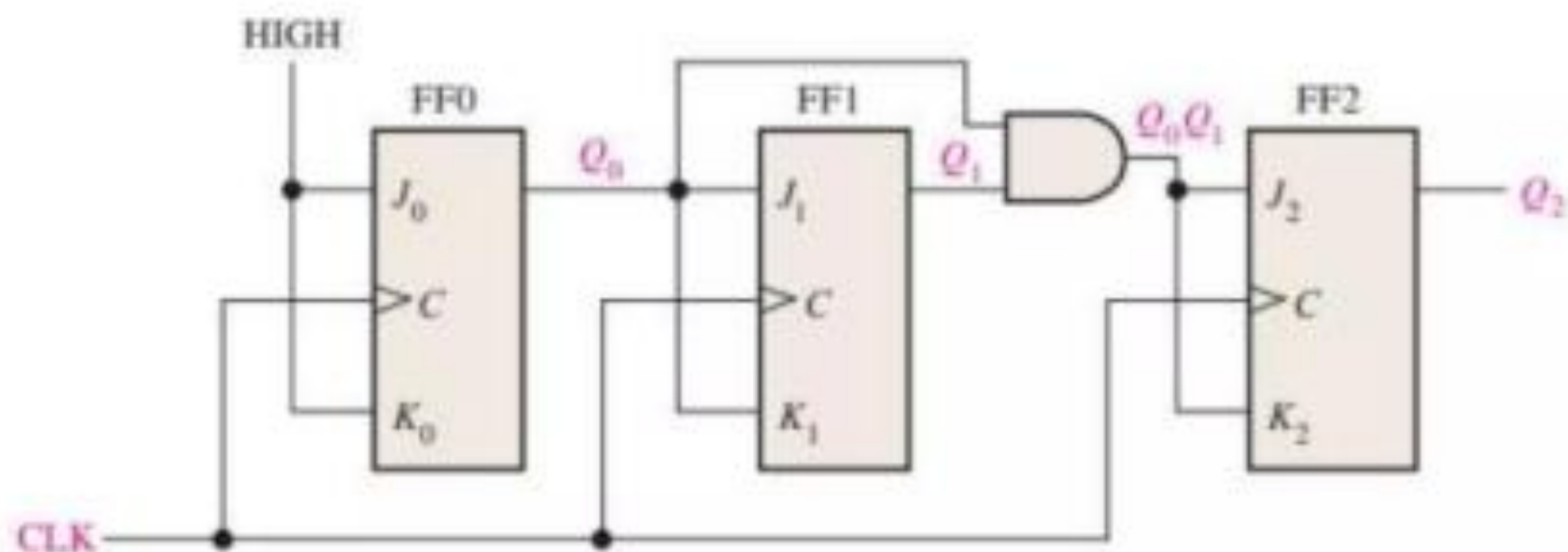
	00	01	11	10
0				
1				

	00	01	11	10
0				
1				

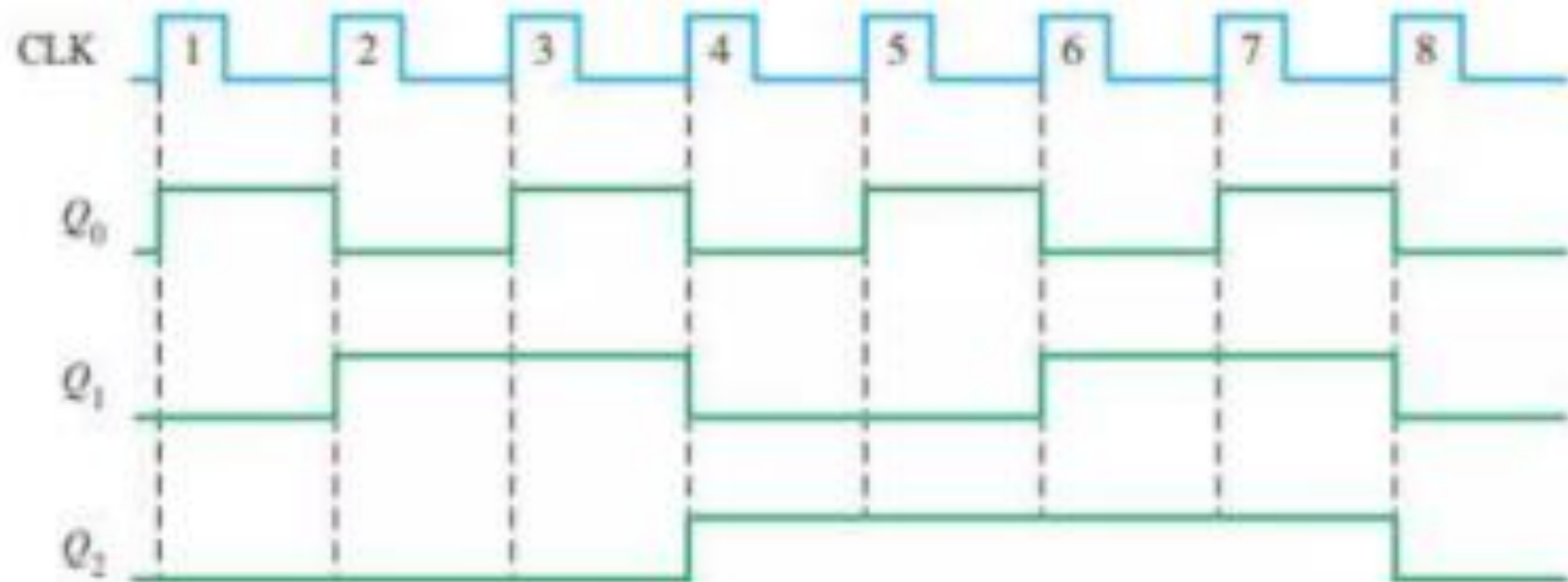
	00	01	11	10
0				
1				

	00	01	11	10
0				
1				

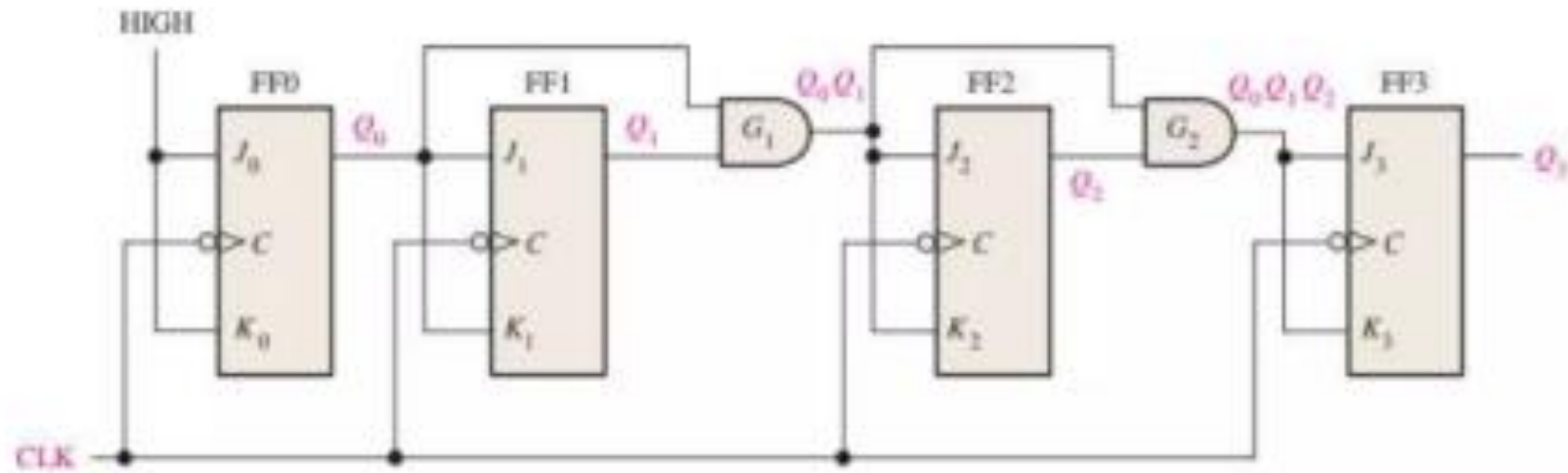
3-Bit Synchronous Binary Counter



Timing Diagram

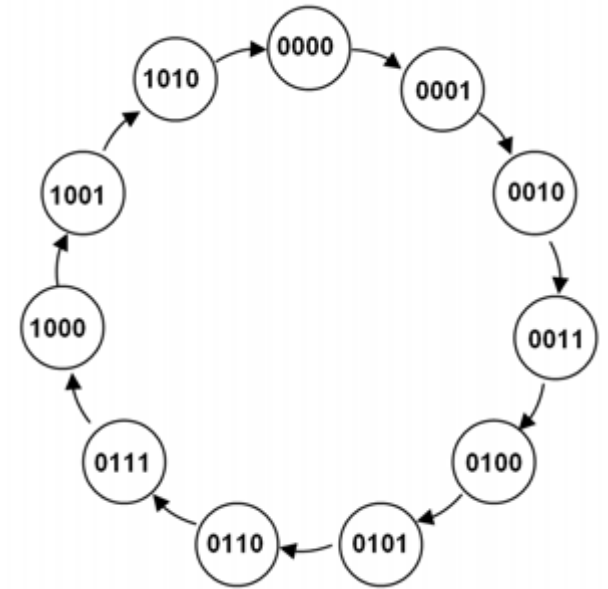


Q: Design a 4-bit Synchronous Counter



4-bit Synchronous Decade Counter

Clock Pulse	Q_3	Q_2	Q_1	Q_0
Initially	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10 (recycles)	0	0	0	0



4-bit Synchronous Decade Counter

present state				Next state				Inputs of Flip-Flop			
Q_3	Q_2	Q_1	Q_0	Q_3	Q_2	Q_1	Q_0	J_{Q_3}, K_{Q_3}	J_{Q_2}, K_{Q_2}	J_{Q_1}, K_{Q_1}	J_{Q_0}, K_{Q_0}
0	0	0	0	0	0	0	1	0 X	0 X	0 X	1 X
0	0	0	1	0	0	1	0	0 X	0 X	1 X	X 1
0	0	1	0	0	0	1	1	0 X	0 X	X 0	1 X
0	0	1	1	0	1	0	0	0 X	1 X	X 1	X 1
0	1	0	0	0	1	0	1	0 X	X 0	0 X	1 X
0	1	0	1	0	1	1	0	0 X	X 0	1 X	X 1
0	1	1	0	0	1	1	1	0 X	X 0	X 0	1 X
0	1	1	1	1	0	0	0	1 X	X 1	X 1	X 1
1	0	0	0	1	0	0	1	X 0	0 X	0 X	1 X
1	0	0	1	0	0	0	0	X 1	0 X	0 X	X 1

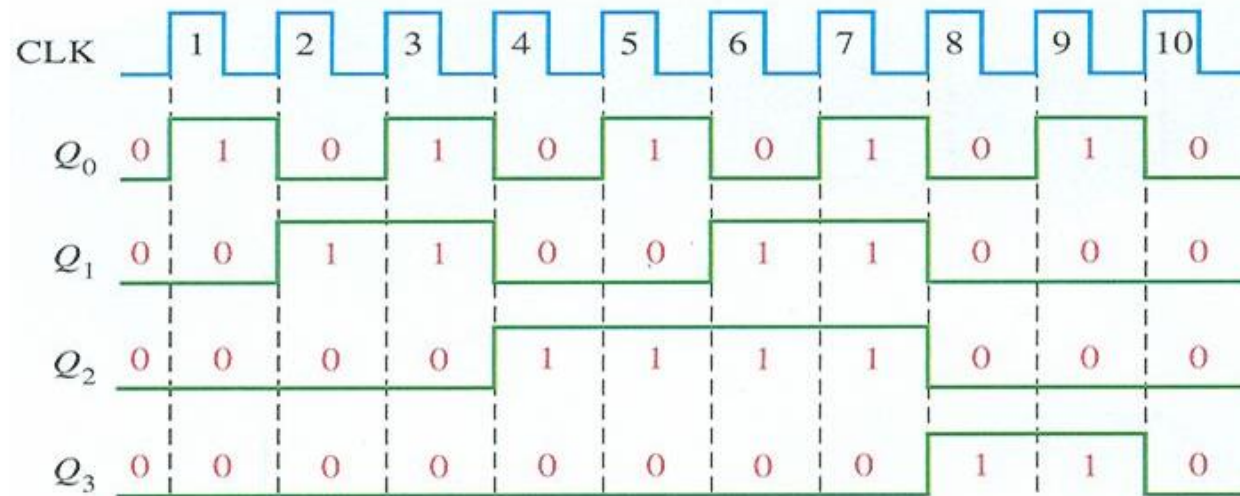
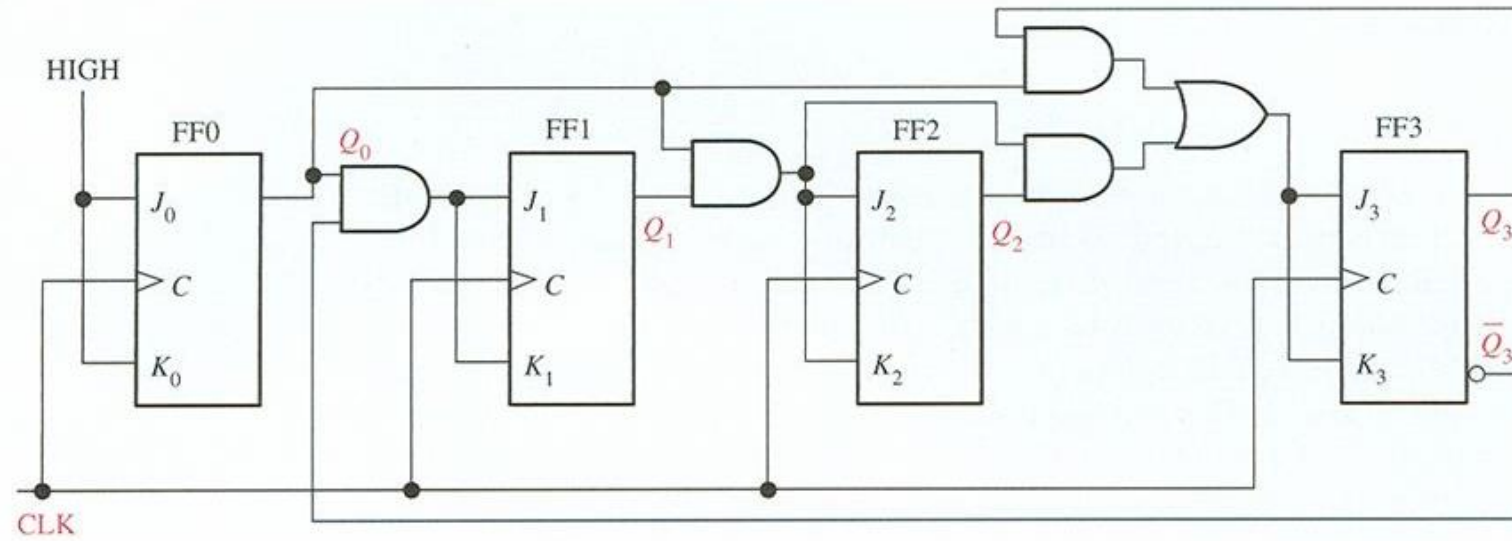
$$J_0 = K_0 = 1$$

$$J_1 = K_1 = Q_0 \bar{Q}_3$$

$$J_2 = K_2 = Q_0 Q_1$$

$$J_3 = K_3 = Q_0 Q_1 Q_2 + Q_0 Q_3$$

4-Bit Synchronous Decade Counter



Modulus of a Counters

The modulus (or MOD number) of a counter refers to the total number of unique states it goes through before repeating the sequence. **Modulus tells you how many counts occur before the counter resets and repeats.**

Examples:

A counter that counts: $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow \text{back to } 0$

It has 4 states that is it is called a MOD-4 counter.

A decade counter like the IC 7490: Counts from 0 to 9

Total states = 10. So it is a MOD-10 counter

General rule:

If a counter has n flip-flops, maximum possible states = 2^n

But actual modulus can be less if designed that way (e.g., MOD-10 using 4 flip-flops instead of MOD-16).

Ring Counters

It consists of n flip-flops, where n is the number of states in the counter. The number of flip-flops determines the number of unique states the counter will cycle through.

One bit in the counter is set to 1, and the rest of the bits are set to 0. In each clock cycle, the 1 bit shifts through the flip-flops, and the pattern repeats.

The output of the last flip-flop is fed back into the first flip-flop, forming a loop, hence the name Ring Counter.

Key Characteristics of a Ring Counter

- Single bit '1': Only one bit in the counter is set to 1 at any given time.
- Cyclic Behavior: The pattern of the bits follows a cyclic behavior and repeats after a fixed number of steps.
- No Reset: The ring counter doesn't automatically reset to zero. Instead, it starts at a defined state and then continues cycling.

Ring Counters

Overriding Input (ORI):

The ORI is used to override the normal functionality of the flip-flops. In this case, Preset (PR) and Clear (CLR) are used as ORI.

- Preset (PR): When the PR signal is 0, the output of the flip-flop (Q) is set to 1. This is an active-low signal.
- Clear (CLR): When the CLR signal is 0, the output of the flip-flop (Q) is set to 0. This is also an active-low signal.

The ORI input of each flip-flop is connected to the Preset (PR) for FF-0 (the first flip-flop) and to Clear (CLR) for FF-1, FF-2, and FF-3 (the other flip-flops).

- At FF-0, when PR = 0, the output Q = 1 is generated, and this creates the initial state of the counter.
- At the other flip-flops (FF-1, FF-2, FF-3), the CLR = 0, forcing the outputs Q = 0 at these flip-flops.

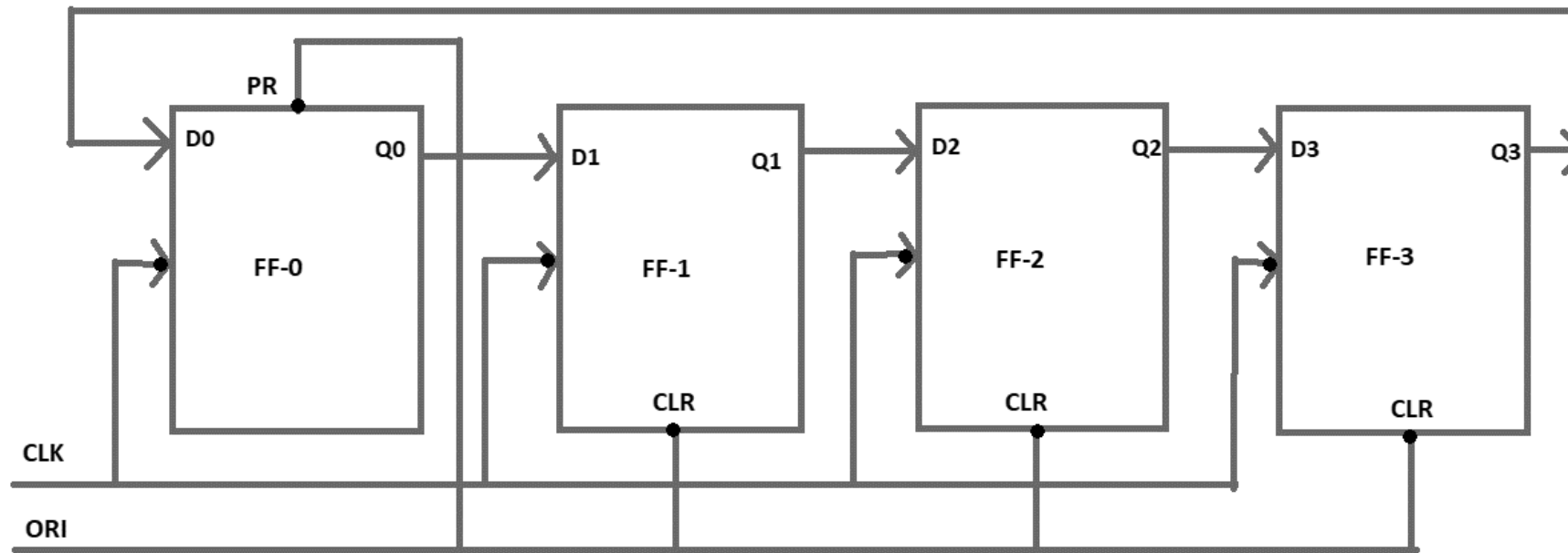
Ring Counters



ORI	CLK	Q0	Q1	Q2	Q3
low	X	1	0	0	0
high	low	0	1	0	0
high	low	0	0	1	0
high	low	0	0	0	1
high	low	1	0	0	0

Straight Ring Counters

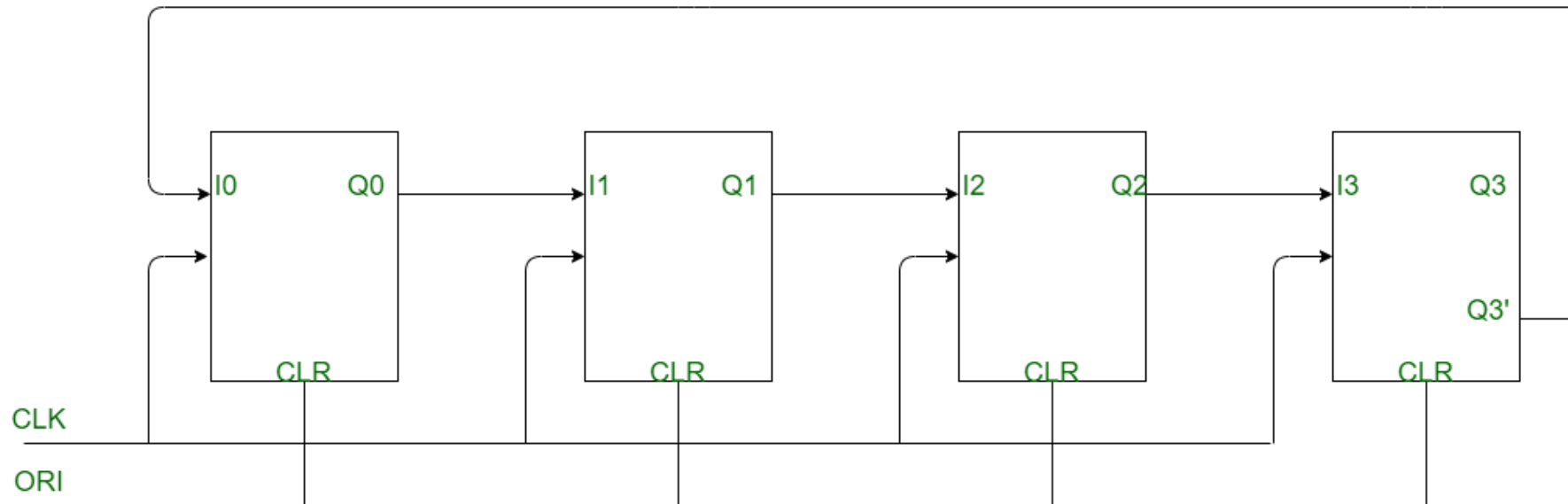
It is also known as One hot Counter. In this counter, the output of the last flip-flop is connected to the input of the first flip-flop. The main point of this Counter is that it circulates a single one (or zero) bit around the ring. Here, we use Preset (PR) in the first flip-flop and Clock (CLK) for the last three flip-flops.



Ring Counter

Twisted Ring/Johnson Counters

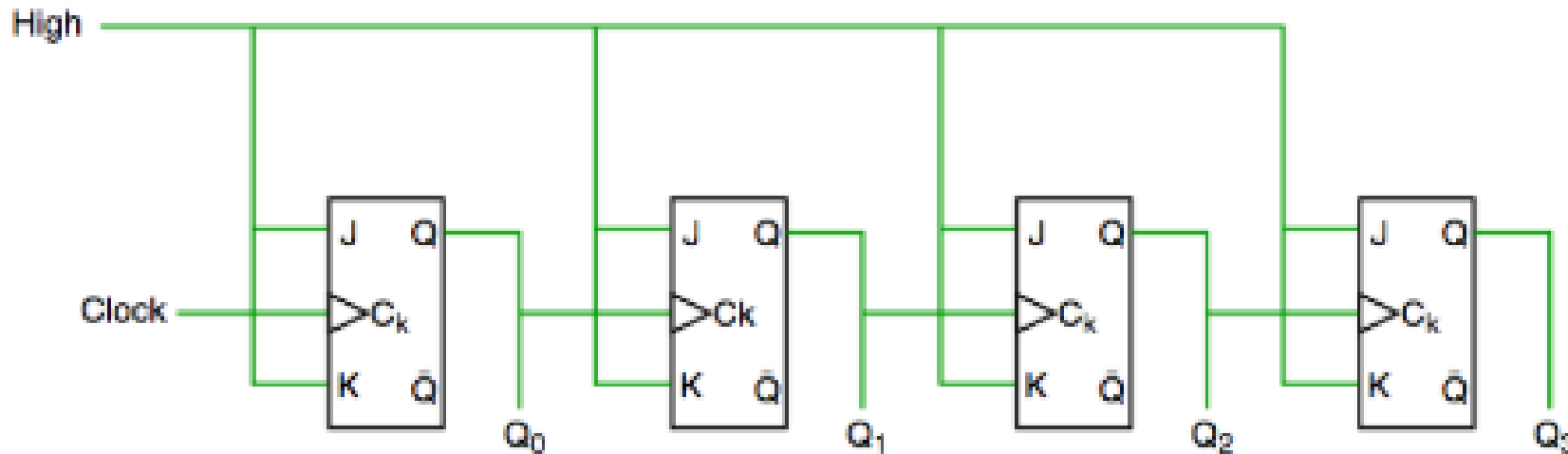
It is also known as a switch-tail ring counter, walking ring counter, or Johnson counter. It connects the complement of the output of the last shift register to the input of the first register and circulates a stream of ones followed by zeros around the ring. Here, we use Clock (CLK) for all the flip-flops. In the Twisted Ring Counter, the number of states = $2 \times$ the number of flip-flops.



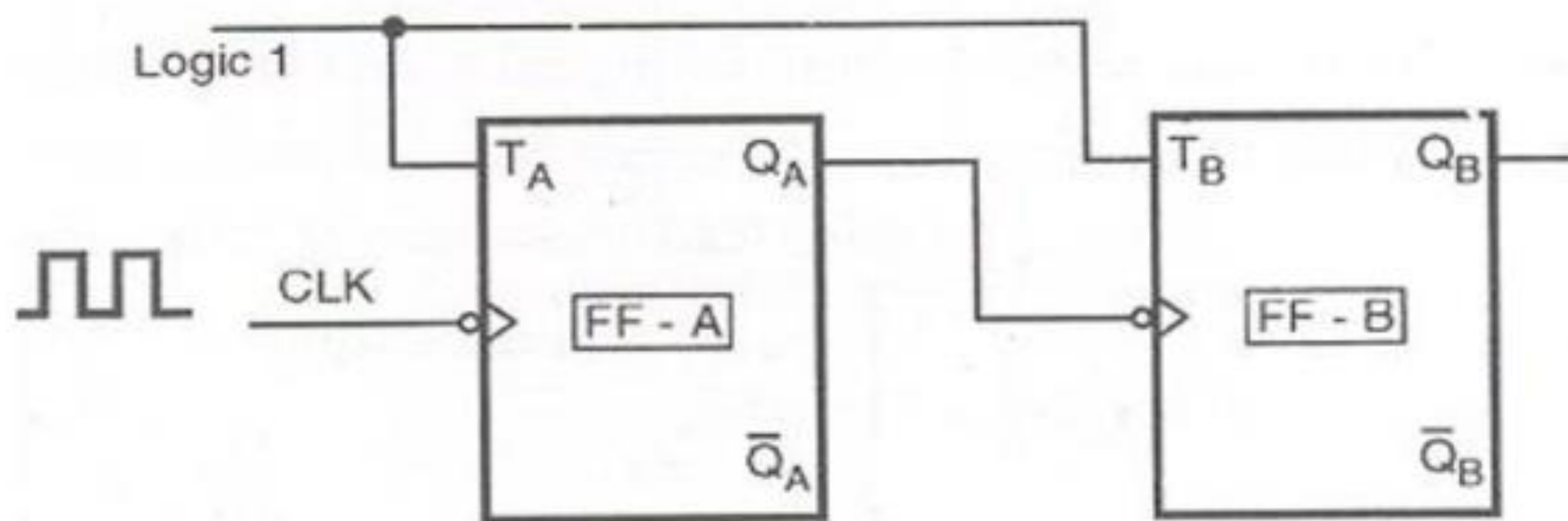
Twisted Ring Counter

Asynchronous Counters

In asynchronous counter we don't use universal clock, only first flip flop is driven by main clock and the clock input of rest of the following flip flop is driven by output of previous flip flops.



2-bit Asynchronous counter Up counter (Ripple Counter)



A two bit asynchronous binary counter

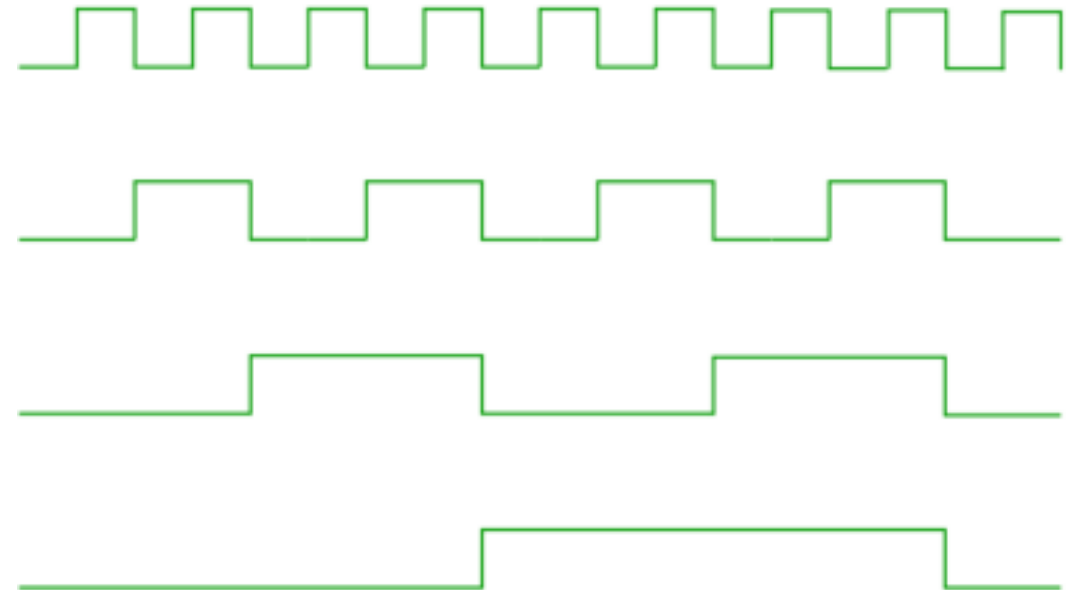
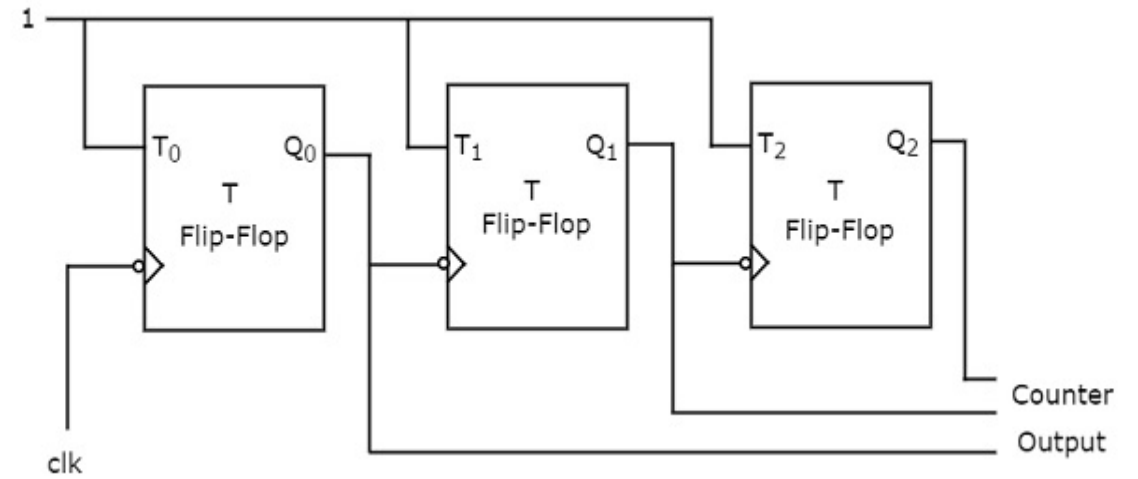
Sr.No	Condition	Operation
1	Initially let both the FFs be in the reset state	$Q_B Q_A = 00$ initially
2	After 1st negative clock edge	As soon as the first negative clock edge is applied, FF-A will toggle and Q_A will be equal to 1. Q_A is connected to clock input of FF-B. Since Q_A has changed from 0 to 1, it is treated as the positive clock edge by FF-B. There is no change in Q_B because FF-B is a negative edge triggered FF. $Q_B Q_A = 01$ after the first clock pulse.
3	After 2nd negative clock edge	On the arrival of second negative clock edge, FF-A toggles again and $Q_A = 0$. The change in Q_A acts as a negative clock edge for FF-B. So it will also toggle, and Q_B will be 1. $Q_B Q_A = 10$ after the second clock pulse.

4	After 3rd negative clock edge	On the arrival of 3rd negative clock edge, FF-A toggles again and Q_A become 1 from 0. Since this is a positive going change, FF-B does not respond to it and remains inactive. So Q_B does not change and continues to be equal to 1. $Q_B Q_A = 11$ after the third clock pulse.
5	After 4th negative clock edge	On the arrival of 4th negative clock edge, FF-A toggles again and Q_A becomes 1 from 0. This negative change in Q_A acts as clock pulse for FF-B. Hence it toggles to change Q_B from 1 to 0. $Q_B Q_A = 00$ after the fourth clock pulse.

Clock	Counter output		State number	Deciimal Counter output
	Q_B	Q_A		
Initially	0	0	—	0
1st	0	1	1	1
2nd	1	0	2	2
3rd	1	1	3	3
4th	0	0	4	0

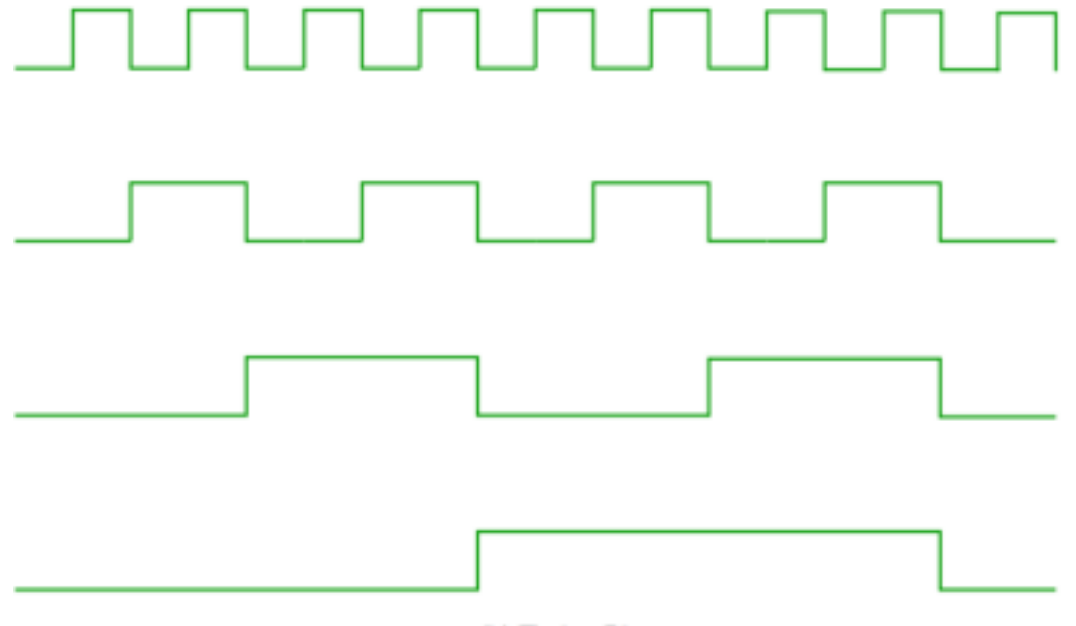
3-bit Up Asynchronous Counters

No of negative edge of Clock	Q_0 <i>LSB</i>	Q_1	Q_2 <i>MSB</i>
0	0	0	0
1	1	0	0
2	0	1	0
3	1	1	0
4	0	0	1
5	1	0	1
6	0	1	1
7	1	1	1



MOD-6 Counters

Maximum count is $2^n - 1$



Thank you!